CNN and Transfer Learning

1. Part 1 - A CNN from Scratch

1. The “From-scratch CNN” gets a test MAE of **13.46 years** (with best validation MAE 11.42 at epoch 19). it’s not really useful because if we look at the age distribution per split we can see that most people are between 20-40 with a long tail up to 90 so the ages have mean around 30 and a spread roughly 15-17 years.

   A model that always predict the training mean would therefore be off by 12 to 13 years on average which is what the CNN basically is doing. Besides, we can see the the first epochs and the final ones have almost the same loss function value which means that the model barely learned more than “predict the average”.
2. In the “From-scratch CNN” plot, train and validation curves stay very close (at the best epoch train MAE = 11.00 vs validation MAE = 11.42), so there is no real overfitting. However, we could argue that the model is underfitting even though it has 406k parameters since the training is so unstable (validation MAE jumps to 15.39 at epoch 5 and 14.07 at epoch 18).

2. Part 2 - Transfer Learning

2.2. By making the fc layer’s parameters training and freezing the rest, we end up with 513 trainable parameters and 11 177 025 other parameters which are frozen. This ratio of approximativelty 0.0046% seems to be logical since ResNet18 only has a relatively small single linear layer at the end for its regression.
2.3. We are using a smaller learning rate (10x smaller) since our model is already trained completely and we only want to achieve small adjustment to make it fit our usecase. Using a higher learning rate would cause overfitting most likely.

3. Experience 1 - Comparing the Three Approaches

Full fine-tuning is clearly the best (8.78 years test MAE), then the frozen ResNet (12.52), then the from-scratch CNN (13.46). This is what CM3 predicts: with a small dataset, starting from pretrained weights beats random initialisation, and adapting the whole network beats keeping it fixed.
1) Unfreezing helped a lot: the test MAE dropped by 3.74 years, and most of the gain came in the first epoch (val MAE 10.58 → 8.45). But the model also started to overfit: at its best epoch the train MAE was 1.75 while the val MAE was 7.11. With 11M parameters and only 4000 images, the network can easily memorise the training faces. What kept it under control here was the low learning rate ($1e^{-4}$), the warm-up of the head in stage 1, and restoring the best epoch.
2) The early layers of a CNN learn generic features like edges, colours and textures, and these are useful for any natural image. Age shows mostly through exactly this kind of cue: wrinkles, skin texture, hair colour, face shape (especially the natural features near the eyes). ImageNet also contains many photos of people. The last layers are more specific to the 1000 object classes, which explains why the frozen model is only slightly better than the from-scratch one.

4. Experience 2 - Does Data Augmentation Help Fine-tuning ?

If we compare the fine-tuned ResNet MAE and the augmented one, we can clearly see an improvement of $8.78 - 8.28 = 0.5$ years which for our usecase is already quite good and significant. This shows us that augmentation can help improving our results.

If we compare the train/validation loss gap from `finetune_history` we can see that the gap is smaller than without augmentation. At the best epoch, the gap is 2.70 vs 7.00 (≈4.3 years) instead of 1.75 vs 7.11 (≈5.4 years), and the final train MSE is 18.3 instead of 10.6, so the model fits the training set less perfectly. This is what we expect from a regulariser: the model sees a slightly different image every epoch, so it is harder to memorise. The gap is still large though, so augmentation reduces overfitting but does not remove it.

While comparing the best validation MAE and test MAE, we can find that: the best validation MAE is almost the same for both runs (7.00 vs 7.11, a difference of only 0.11) while the test MAE differs by 0.50. Looking only at validation, We could have concluded that augmentation does nothing. With 1200 images and errors that vary a lot from one face to another, the validation MAE has a noise of a few tenths of a year, so differences of 0.1–0.5 years are close to that noise. Furthermore, each configuration was trained only once. So validation MAE alone

is not a fully reliable signal for small changes but one solution could be to get a bigger validation dataset with a wide array of images.

5. Experience 3 - How Much of the Network Should You Fine-tune ?

If we take a look at the “All Five Approaches Compared”, we can find that: partial fine-tuning (layer 4.1 + fc) reaches 10.74 years MAE, exactly between frozen (12.52 years MAE) and full fine-tuning (8.78 years MAE).
This shows us that going from frozen to partial gains 1.78 years and from partial fine-tuning to full fine tuning gains another 1.96 years.

Partial fine-tuning trains 4.72M parameters ($\approx$ 42% of the possible parameters) instead of 11.18M, but it loses about 2 years of MAE compared to full fine-tuning. It also overfits almost as much (train MAE $\approx$ 2.3 vs validation $\approx$ 9.0 in “Partial Fine-tuning (layer4.1 + fc only)”). As it seems, age depends a lot on fine texture details encoded in the earlier and middle layers, and those stay fixed in partial fine-tuning. With 4000 images and a GPU, the extra cost of full fine-tuning is small, so here the trade-off is not entirely worth it (and needless to say on a CPU that takes tens of minutes to execute a single stage, this is definitely not worth it).

With only 500 images: we would expect full fine-tuning to look relatively worse compared to partial fine-tuning. With 11M free parameters and 500 examples, the model would overfit much faster and could also “forget” its good ImageNet features. Freezing the early layers acts as a strong constraint, so partial fine-tuning (or even frozen) would probably be the better choice.

6. Overall comparison

1) If we rank the five models based on their gain we obtain this ranking: Full fine-tuning + augmentation (8.28) < full fine-tuning (8.78) < partial fine-tuning (10.74) < frozen (12.52) < from scratch (13.46).
We can see that the augmentation only adds 0.5 years of improvement which is small compared to the 3.74 years gained by going from frozen to fine-tuned so the decision that matters most is fine-tuning the pretrained network.
2) if we look at the best epoch column we can see that almost all models got their best validation MAE almost at the end of the 20 epochs (scratch 19, partial 20, full fine-tuning 19, augmentation 18) so they were still improving a bit and a few more epochs could probably help. the frozen model is different, its best epoch is 16 but its validation MAE was already stuck around 10.6-10.8 since epoch 12 and it makes sense because it only trains 513 parameters so it learns fast and then stops improving. for the fine-tuned models we notice that most of the improvement happens in the first 3-4 epochs.
3) When it comes to the best accuracy per training cost we can clearly identify that the full fine-tuning with data augmentation model is the best solution for this dataset, it gets the lowest test MAE and trains the same number of parameters (11.18M) as normal full fine-tuning and augmentation doesn’t add any parameter so the extra 0.5 years of improvement comes for free. Partial fine-tuning saves 58% of the trainable parameters but is 2.5 years worse and the frozen model is the cheapest (513 parameters) but it has the worst performance of the ResNet models (12.52). As such, a full fine-tuning with data augmentation is clearly the best choice.

7. Experience 4 - Feature Map Visualisation

Each channel picks up something different if we look at the “layer1 feature maps” figure.
We can see in the top row that in the fifth map the whole face lights up, so we can almost see the face shape and the mouth. It seems to react to bright, smooth skin areas.
In the sixth map, we notice that it is dark almost everywhere except for thin curved lines along the outline of the face and the hair, so it looks like it detects edges.
Other channels look more like texture detectors. so what we can notice is that in general, early layers detect simple, low-level features like edges, bright regions and textures, and because these features are not specific to one usecase of IA , they transfer well from ImageNet.